Deep Neural Network
Module # 3 : Linear Neural Networks for Regression

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Work Integrated Learning Programmes

The instructor is gratefully acknowledging
the authors who made their course
materials freely available online.
This deck is prepared by Seetha Parameswaran.

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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What is Regression?

Definition

Regression is a supervised learning task that predicts a continuous-valued output based on input features.

Key Characteristics:

- Input: Feature vector $\mathbf{x} \in \mathbb{R}^d$
- Output: Real-valued target $y \in \mathbb{R}$
- Goal: Learn a function $f : \mathbb{R}^d \to \mathbb{R}$ such that $y \approx f(\mathbf{x})$

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Examples of Regression Problems

1. House Price Prediction
   - Input: Size, location, number of rooms
   - Output: Price in dollars
2. Temperature Forecasting
   - Input: Historical weather data, season, humidity
   - Output: Temperature in degrees
3. Stock Price Prediction
   - Input: Historical prices, trading volume, market indicators
   - Output: Future stock price

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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What is Linear Regression?

Definition

Linear regression assumes the output is a linear combination of the input features.

Mathematical Form:

$$y = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_d x_d \tag{1}$$

where:

- $\mathbf{x} = [x_1, x_2, \ldots, x_d]^T$ are the input features
- $\mathbf{w} = [w_1, w_2, \ldots, w_d]^T$ are the weights
- w_0 is the bias term

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Examples of Linear Regression

1. Salary Prediction
   - Salary $= w_0 + w_1 \times$ Years of Experience
2. Crop Yield Prediction
   - Yield $= w_0 + w_1 \times$ Rainfall $+ w_2 \times$ Fertilizer
3. Energy Consumption
   - Energy $= w_0 + w_1 \times$ Temperature $+ w_2 \times$ Hour $+ w_3 \times$ Day

The relationship between inputs and output is assumed to be linear.

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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THE FOUR COMPONENTS

A complete machine learning system consists of:

1. Data: d-dimensional input vectors and target outputs
2. Model: Single neuron implementing linear function
3. Objective Function: Measures prediction error
4. Learning Algorithm: Gradient descent to optimize weights

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Matrix Form of Data

For efficient computation, we organize data into matrices:

Input Data = Augmented Features:

Add bias term: $\tilde{\mathbf{x}}^{(i)} = [1, x_1^{(i)}, \ldots, x_d^{(i)}]^T$

Each row = one training example

Design Matrix: $\mathbf{X} \in \mathbb{R}^{N \times (d+1)}$ $\quad \mathbf{X} = \begin{bmatrix} 1 & x_1^{(1)} & \cdots & x_d^{(1)} \\ 1 & x_1^{(2)} & \cdots & x_d^{(2)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(N)} & \cdots & x_d^{(N)} \end{bmatrix}$

Target Vector: $\mathbf{y} \in \mathbb{R}^N$ $\quad \mathbf{y} = \begin{bmatrix} y^{(1)} & y^{(2)} & \ldots & y^{(N)} \end{bmatrix}^\top$

Weight Vector: $\mathbf{w} \in \mathbb{R}^{d+1}$ $\quad \mathbf{w} = \begin{bmatrix} w_0 & w_1 & w_2 & \ldots & w_d \end{bmatrix}^\top$

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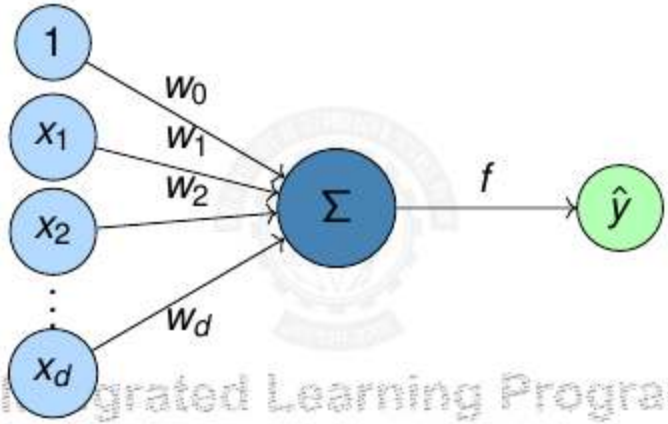
Linear Model as a Single Neuron

Model linear regression as a single artificial neuron:



- Input: d-dimensional feature vector $\mathbf{x} \in \mathbb{R}^d$
- Output: Single predicted value $\hat{y} \in \mathbb{R}$

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LINEAR MODEL PREDICTION

Model: A single neuron computes a weighted sum and applied identity activation function.

Prediction for input $\mathbf{x}$:

$$\hat{y} = f(\mathbf{w}^T\mathbf{x}) = f(w_0 + w_1x_1 + w_2x_2 + \cdots + w_dx_d) \tag{2}$$

where:

- $\mathbf{w} = [w_0, w_1, \ldots, w_d]^T \in \mathbb{R}^{d+1}$ are the model parameters
- $\mathbf{x} = [1, x_1, \ldots, x_d]^T \in \mathbb{R}^{d+1}$ is the input
- $f(\cdot)$ is the identity activation $f(z) = z$

Vectorized form: for all predictions

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w} \tag{3}$$

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Identity Activation Function

Identity Activation Function

$$f(z) = z \tag{4}$$

Properties of Identity Function

- Output: Continuous, $(-\infty, \infty)$
- Differentiable everywhere: $f'(z) = 1$
- Allows output to take any real value
- Gradient flows unchanged through the neuron
- Perfect for predicting continuous targets

Why identity for regression?

- Target values are continuous (prices, temperatures, etc.)
- Need to predict any real number, not just binary $\{ 0,1 \}$
- Differentiability enables gradient-based optimization
- No information loss from input to output

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Objective Function: Squared Error Loss

We need to measure how well our model fits the data.

Loss for single example:

$$\ell\left(\mathbf{w}; \mathbf{x}^{(i)}, y^{(i)}\right) = \frac{1}{2}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \frac{1}{2}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)^2 \tag{5}$$

Total Loss (Mean Squared Error):

$$J(\mathbf{w}) = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{2}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)^2 \tag{6}$$

Goal: Find $\mathbf{w}^*$ that minimizes $J(\mathbf{w})$

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} J(\mathbf{w}) \tag{7}$$

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Why Squared Error?

Advantages of squared error loss:

- Differentiable: Smooth everywhere, easy to optimize
- Convex: Single global minimum (for linear models)
- Penalizes large errors: Quadratic growth
- Statistical interpretation: Maximum likelihood under Gaussian noise

Vector Form:

$$J(\mathbf{w}) = \frac{1}{2N}\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 = \frac{1}{2N}(\mathbf{X}\mathbf{w} - \mathbf{y})^T(\mathbf{X}\mathbf{w} - \mathbf{y}) \tag{8}$$

where $\mathbf{X} \in \mathbb{R}^{N\times(d+1)}$ is the design matrix and $\mathbf{y} \in \mathbb{R}^N$ is the target vector.

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Batch Gradient Descent Algorithm

Algorithm 1: Gradient Descent for Linear Regression

Input: Dataset $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{N}$, learning rate η, tolerance ϵ

Output: Learned weights $\mathbf{w}$

```
Initialize w^(0) = 0 (or random);
t ← 0;
while not converged do
    Compute gradient: ∇J(w^(t)) = (1/N) X^T (X w^(t) − y);
    Update weights: w^(t+1) = w^(t) − η ∇J(w^(t));
    if ||∇J(w^(t+1))|| < ε then
        break // Converged
    t ← t + 1;
return w^(t+1);
```


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Computing the Gradient

Gradient of squared error loss:

$$\nabla J(\mathbf{w}) = \frac{\partial J}{\partial \mathbf{w}} = \frac{1}{N}\sum_{i=1}^{N}\left(\mathbf{w}^T\mathbf{x}^{(i)} - y^{(i)}\right)\mathbf{x}^{(i)} \tag{9}$$

Matrix Form:

$$\nabla J(\mathbf{w}) = \frac{1}{N}\mathbf{X}^T\left(\mathbf{X}\mathbf{w} - \mathbf{y}\right) \tag{10}$$

where $\mathbf{X}^T \in \mathbb{R}^{(d+1)\times N}$, $\mathbf{X}\mathbf{w} - \mathbf{y} \in \mathbb{R}^N$, so $\nabla J \in \mathbb{R}^{d+1}$

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Gradient Descent Update Rule

Update equation:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \frac{\eta}{N}\mathbf{X}^T\left(\mathbf{X}\mathbf{w}^{(t)} - \mathbf{y}\right) \tag{11}$$

Key Parameters:

- Learning rate η: Controls step size
  - Too large: May overshoot minimum
  - Too small: Slow convergence
- Stopping criterion: $\|\nabla J(\mathbf{w})\| < \epsilon$ or max iterations

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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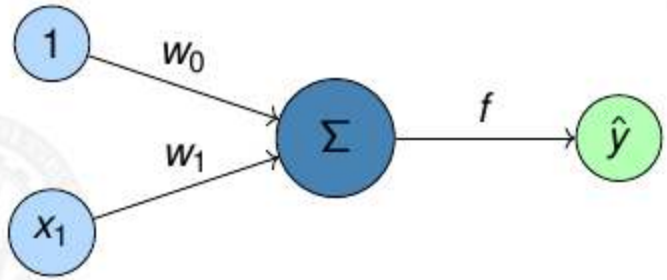
Numerical Example: Setup

Tiny Dataset: Predict house price from size

Size (x_1)	Price (y)
1	2
2	4
3	5



Matrix Formulation:

$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 2}, \quad \mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} \in \mathbb{R}^2$$

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Numerical Example: Initialization

Hyperparameters: $\eta = 0.1 \quad N = 3$

Initialize weights: $\mathbf{w}^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Vectorized Prediction: $\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}$

Initial predictions: $\hat{\mathbf{y}}^{(0)} = \mathbf{X}\mathbf{w}^{(0)} = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Loss Function: $J(\mathbf{w}) = \frac{1}{2N} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

Initial loss: $J(\mathbf{w}^{(0)}) = \frac{1}{6} \left\| \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} \right\|^2 = \frac{1}{6}(4 + 16 + 25) = 7.5$

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Numerical Example: Computing Gradient

Error vector: $\mathbf{e}^{(0)} = \hat{\mathbf{y}}^{(0)} - \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \\ -5 \end{bmatrix}$

Gradient computation: $\nabla J(\mathbf{w}^{(0)}) = \frac{1}{N}\mathbf{X}^T\mathbf{e}^{(0)}$

$$\nabla J(\mathbf{w}^{(0)}) = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}\begin{bmatrix} -2 \\ -4 \\ -5 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} -2-4-5 \\ -2-8-15 \end{bmatrix} = \begin{bmatrix} -3.67 \\ -8.33 \end{bmatrix}$$

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Numerical Example: Weight Update

Gradient descent update: $\mathbf{w}^{(1)} = \mathbf{w}^{(0)} - \eta \nabla J(\mathbf{w}^{(0)})$

Computing new weights: $\mathbf{w}^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - 0.1 \begin{bmatrix} -3.67 \\ -8.33 \end{bmatrix} = \begin{bmatrix} 0.367 \\ 0.833 \end{bmatrix}$

New predictions: $\hat{\mathbf{y}}^{(1)} = \mathbf{X}\mathbf{w}^{(1)}$

$$\hat{\mathbf{y}}^{(1)} = \begin{bmatrix} 0.367 & 0.833 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1.20 \\ 2.03 \\ 2.87 \end{bmatrix}$$

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Numerical Example: New Loss

New error vector: $\mathbf{e}^{(1)} = \hat{\mathbf{y}}^{(1)} - \mathbf{y} = \begin{bmatrix} 1.20 \\ 2.03 \\ 2.87 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} -0.80 \\ -1.97 \\ -2.13 \end{bmatrix}$

Loss after one iteration: $J(\mathbf{w}^{(1)}) = \frac{1}{6}\|\mathbf{e}^{(1)}\|^2 = \frac{1}{6}(0.64 + 3.88 + 4.54) \approx 1.51$

Progress:

- Initial loss: $J(\mathbf{w}^{(0)}) = 7.5$
- After 1 iteration: $J(\mathbf{w}^{(1)}) = 1.51$
- **80% reduction!** ✓

Continue until $\|\nabla J(\mathbf{w})\| < \epsilon$ or max iterations reached

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Computational Graph for One Gradient Descent Step

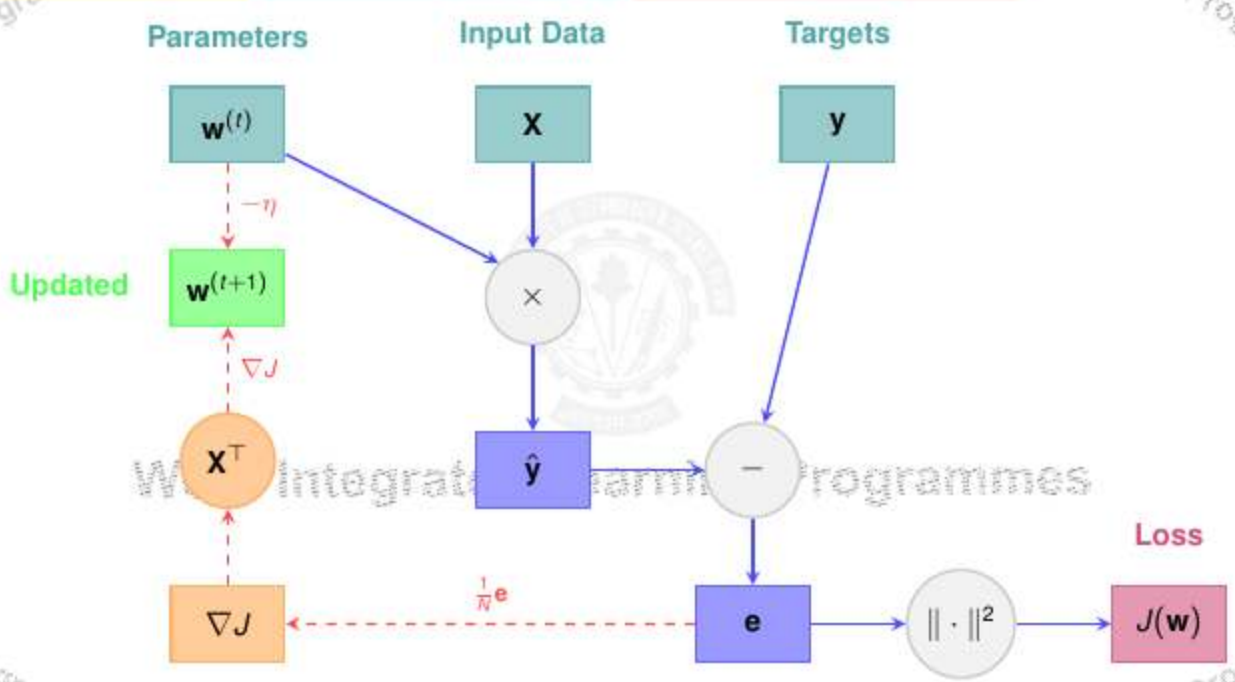

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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Model Evaluation: Training vs Test Error

Key Principle: Evaluate on unseen data

Data Split:

- Training set: Used to learn weights $\mathbf{w}$ (typically 90-99%)
- Test set: Used to evaluate generalization (typically 1-10%)

$$\text{Training Error:} \quad J_{\text{train}} = \frac{1}{N_{\text{train}}} \sum_{i \in \text{train}} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

$$\text{Test Error:} \quad J_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i \in \text{test}} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

Goal: Minimize test error (generalization performance)

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EVALUATION METRICS

1. Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 \tag{12}$$

2. Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\text{MSE}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}^{(i)} - y^{(i)}\right)^2} \tag{13}$$

3. Mean Absolute Error (MAE):(Less sensitive to outliers than MSE)

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N} \mid y^{(i)} - \hat{y}^{(i)} \mid \tag{14}$$

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R^2 Score (Coefficient of Determination)

Definition: Proportion of variance explained by the model

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y^{(i)} - \hat{y}^{(i)})^2}{\sum_{i=1}^{N}(y^{(i)} - \bar{y})^2} = 1 - \frac{SS_{\text{residual}}}{SS_{\text{total}}} \tag{15}$$

where $\bar{y} = \frac{1}{N}\sum_{i=1}^{N} y^{(i)}$ is the mean.

Interpretation:

- $R^2 = 1$: Perfect fit (all variance explained)
- $R^2 = 0$: Model no better than predicting mean
- $R^2 < 0$: Model worse than baseline (possible on test set)
- Typical: $0.7 < R^2 < 0.9$ indicates good fit

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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Tips for Implementation

1. Choosing Learning Rate η:
   - Start with $\eta \in \{0.001, 0.01, 0.1, 1.0\}$
   - Plot loss vs iterations for each
   - If loss increases or becomes NaN $\rightarrow$ decrease η
   - If convergence too slow $\rightarrow$ increase η
   - Use learning rate schedules for fine-tuning
2. When to Stop Training:
   - Gradient magnitude: $\|\nabla J\| < \epsilon$ (e.g., $\epsilon = 10^{-4}$)
   - Loss change: $|J^{(t)} - J^{(t-1)}| < \epsilon$
   - Maximum iterations reached (e.g., $T = 1000$ or 10,000)
   - Validation loss starts increasing (early stopping for overfitting)

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More Practical Tips

3. Weight Initialization:
   - Zero initialization: $\mathbf{w}^{(0)} = \mathbf{0}$ (works well for linear regression)
   - Avoid large initial values (can cause divergence or overflow)

4. Feature Engineering:
   - Domain knowledge: Create meaningful derived features (e.g., price per sq ft)
   - Feature selection: Remove irrelevant/redundant features

5. Feature Scaling and Normalization:
   - Features with different scales can slow convergence
   - 1. Min-Max Scaling: Scale to $[0, 1]$ $x_j' = \frac{x_j - \min(x_j)}{\max(x_j) - \min(x_j)}$
   - 2. Standardization (Z-score): Zero mean, unit variance $x_j' = \frac{x_j - \mu_j}{\sigma_j}$

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Debugging Checklist

Common Issues and Solutions:

- Loss is NaN/Inf:
  - Learning rate too large
  - Numerical overflow (check feature scales)
  - Solution: Decrease η, scale features
- Training works but test error high:
  - Overfitting (too many features relative to data)
  - Solution: Regularization, more data, or simpler model
- Loss oscillating:
  - Learning rate too large
  - Solution: Decrease η or use learning rate decay
- Loss not decreasing:
  - Learning rate too small
  - Bug in gradient computation (check signs!)
  - Wrong sign in update rule (should be minus)
  - Solution: Increase η, verify code

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Table of Contents

1. Regression
2. Linear Regression
3. The Linear Model: Single Neuron
4. Worked Example
5. Evaluation
6. Implementation Tips
7. Summary

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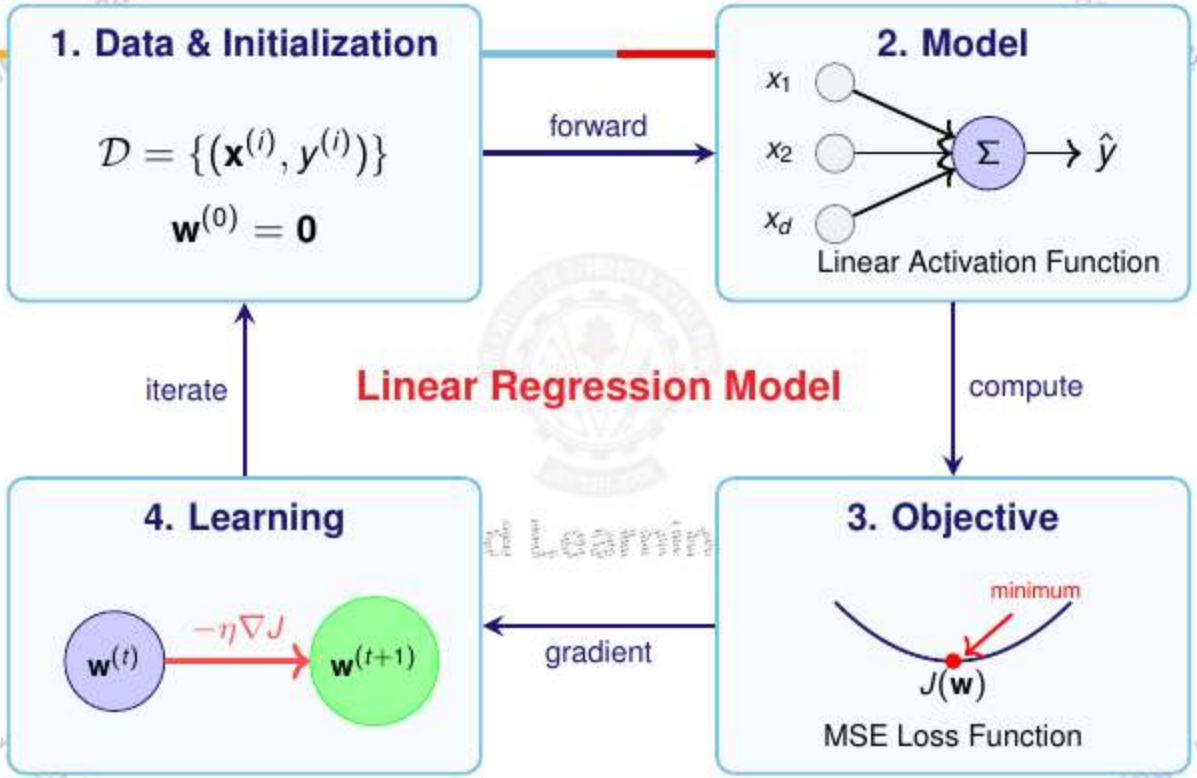
SUMMARY: PUTTING IT ALL TOGETHER



This framework extends to more complex models: neural networks, deep learning!

Key Takeaways

- Regression predicts continuous outputs from input features.
- Linear regression assumes a linear relationship: $y = \mathbf{w}^T\mathbf{x} + w_0$
- A single neuron with identity activation implements linear regression.
- Squared error loss provides a convex objective function.
- Gradient descent iteratively optimizes weights: $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla J$
- Feature scaling is critical for fast convergence.
- Evaluation on test set ensures generalization.

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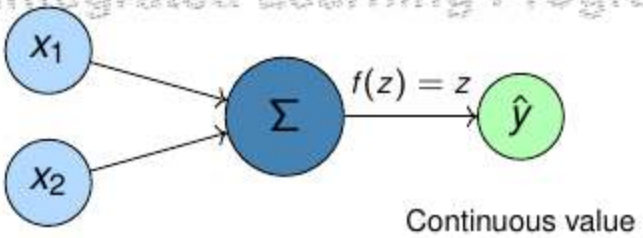
LINEAR REGRESSION

Key insight: Linear relationship between inputs and output, optimized by minimizing squared errors.

- Problem: Predict continuous values: $y \in \mathbb{R}$
- Model: Single neuron with identity activation
- Output: Predicted value $\hat{y} = \mathbf{w}^T\mathbf{x} \in (-\infty, \infty)$
- Loss: Mean Squared Error (MSE)
- Training: Batch Gradient Descent (GD)



Thank you :)
All the best :)

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